

 **DEFINITION OF AUTOMATION:** Automation is defined as a technology concerned with the application of Mechanical, Electronics, and Computer-based systems to operate and control production.

This technology includes:

- Automated machines
- Automated assembly systems
- Automated inspection systems for quality control
- Industrial robots

Examples of industries using above types of systems are :

- Automobile industries
- Aircraft industries
- Metal forming industries
- Chemical industries

ADVANTAGES OR THE MAIN OBJECTIVES OF AUTOMATION ARE

- To increase production rate
- To improve the product quality
- To reduces operation time
- To reduce production cost
- To provide safer working conditions for the workers
- Replacing human operators in tasks that involve hard physical work
- Replacing humans operators in tasks that involve dangerous environments (i.e. fire, space, chemicals, nuclear industries)

THE MAIN DISADVANTAGES OF AUTOMATION ARE:

- Automation results in unemployment (worker displacement)
- Automation reduces the need of skilled labour
- High initial cost
- Benefit suggestions from employees is lost(because of worker displacement)

TYPES OF AUTOMATION

1. *Fixed Automation*
2. *Programmable Automation*
3. *Flexible Automation*

1. Fixed Automation:

- Fixed automation, also known as "hard automation,"
- Fixed automation is a system in which the sequence of processing operations is fixed by the equipment configuration(shape or design)
- In this type of automation, equipment is specially designed to produce a particular type of product
- If the product changes, the same equipments cannot be used
- It is used when volume of production is high and product variety is very low
- Examples of fixed automation are automated production line machining transfer lines, automated assembly machines,

2. Programmable Automation:

- In programmable automation, the production equipment is designed with the capability to change the sequence of operations to accommodate different product configurations.

- The sequence of operations are controlled by a program of instructions
- If the product is changed, only program of instructions are changed but not the equipment
- New programs can be prepared and entered into the equipment to produce new products.
- This kind of automation is suitable for Batch production
- To produce each new batch, the production equipment must be reprogrammed to accommodate the new product style. This changeover procedure takes time.
- Examples of programmable automation are numerically controlled (NC) machine tools and industrial robot.

3. Flexible Automation:

- It is an extension of programmable automation
- In this type of automation, the system is capable of changing over from one job to the next with no lost time between jobs
- That is, equipment produces a variety of parts or products with no lost time between jobs
- This type of automation is used where product varieties are high and production volumes are low

Examples of flexible automation are the flexible manufacturing systems

Note: Flexible means Capable of being changed

DEFINITION OF AN INDUSTRIAL ROBOT:

An industrial robot is a general-purpose programmable machine having certain anthropomorphic characteristics i.e. human-like characteristics.

The most anthropomorphic, or human-like, characteristics are

1. Robot's mechanical arm. The mechanical arm is used to perform the various industrial tasks.
2. Sensors to respond to input
3. Intelligence to make decisions

The above capabilities permit Robots to perform a variety of useful tasks

OR

Defined by Robotics Industry Association (RIA) as

An industrial robot is a programmable, multifunctional manipulator designed to move materials, parts, tools and special devices through variable programmed motions for performance of variety of tasks.

ROBOT ANATOMY OR STRUCTURE OF INDUSTRIAL ROBOT

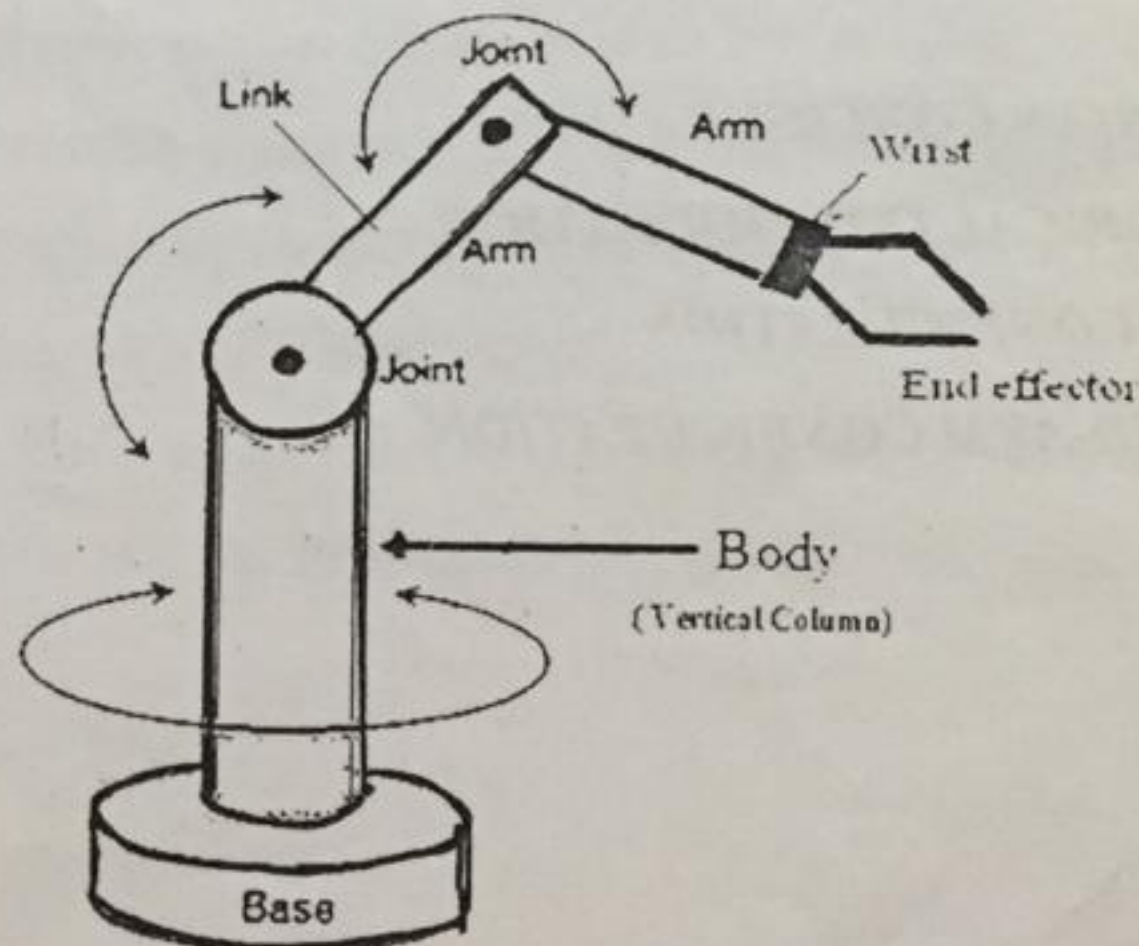


Fig: Robot Anatomy Or Structure Of Industrial Robot

- Fig shows robot's anatomy
- Robot anatomy is concerned with the physical construction of the body, arm and wrist of the machine
- The robots are mounted on the base. Body of the robot is attached to the base
- Arm assembly of the robot is attached to the body
- Arm assembly consists of series of links and joints
- Links are rigid members between joints
- Joint provides relative motion between two links of the robot
- At the end of the arm, wrist is attached
- The body, arm and wrist is called as Manipulator
- A hand/tool is attached at the end of the wrist and is called as end effector
- End effectors are grippers or tools which perform material handling, welding, painting or any other function

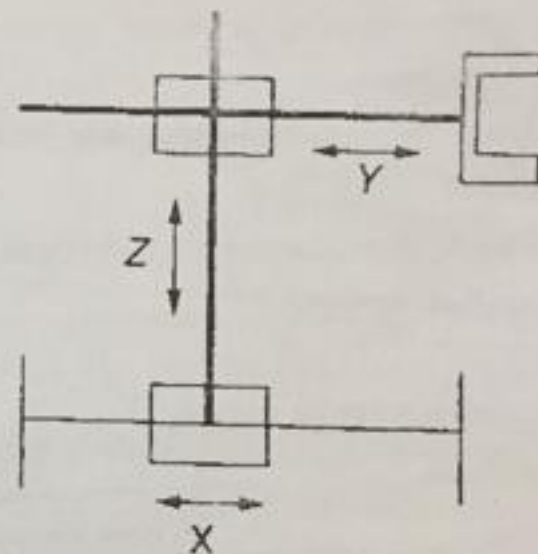
CLASSIFICATION OF ROBOTS:

Classification based on Physical Configurations

- CARTESIAN CONFIGURATION
- CYLINDRICAL CONFIGURATION
- POLAR CONFIGURATION
- JOINTED-ARM CONFIGURATION

CLASSIFICATION BASED ON PHYSICAL CONFIGURATIONS

3. Cartesian Configuration



- Fig shows Cartesian configuration robot
- It consists of three perpendicular slides. The three slides move in X, Y and Z direction and hence it is called the Cartesian co-ordinate robots
- This robot can move up and down, left and right, backward and forward, but it has no rotation
- These robots are used for pick and place operations like material handling and loading and unloading of work pieces in machines
- Motions of this kind of robot traces rectangular work space

4. Cylindrical Configuration:

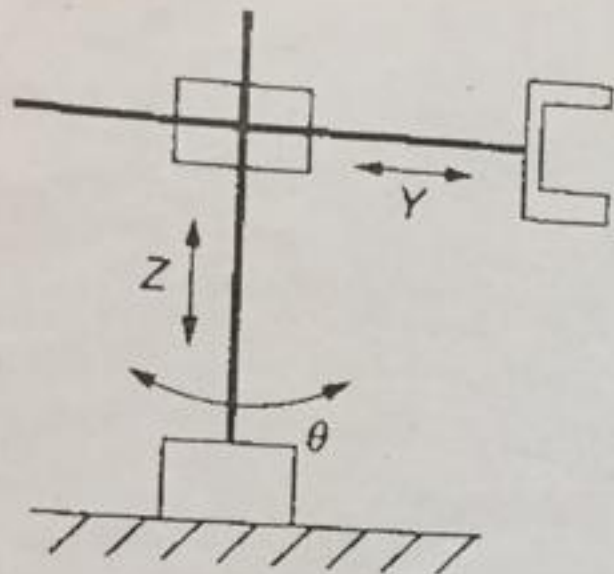
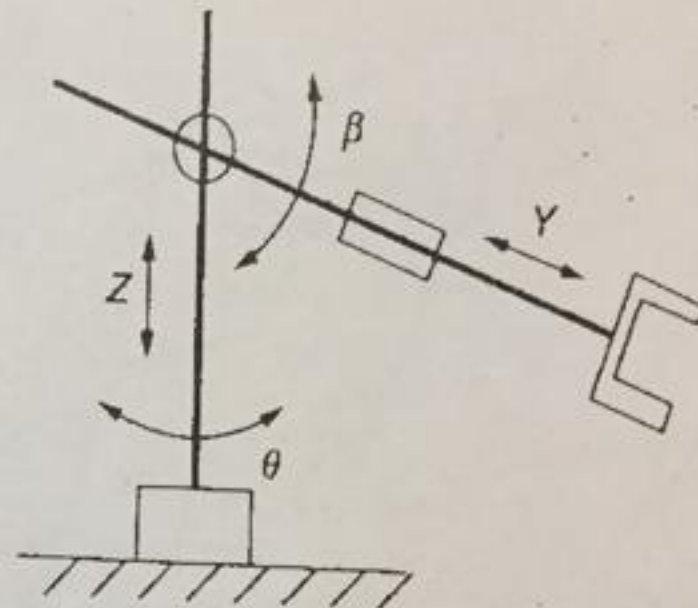


Fig: Cylindrical Configuration

- Fig shows Cylindrical Configuration robot
- It consists of vertical column, a slide and robot arm which is attached to the slide
- The vertical column is mounted on a base and that swivels about the base
- Slide which moves up and down along the vertical column
- Robot arm moves radially (in and out) with respect to vertical column.
- Motions of this kind of robot traces Cylinder work space and hence this configuration is also called as Cylindrical Configuration robot

5. Polar Configuration:



- Fig shows Polar configuration robot. It is also called as spherical coordinate robot
- It consists of base, vertical column sliding arm & Pivot
- The vertical column is mounted on a base and that swivels about the base
- The arm extends and retracts to provide reach as shown in fig
- Pivot is used to raise and lower the arm
- Motions of this kind of robot traces partial sphere work shape and hence this configuration is also called as spherical Configuration robot

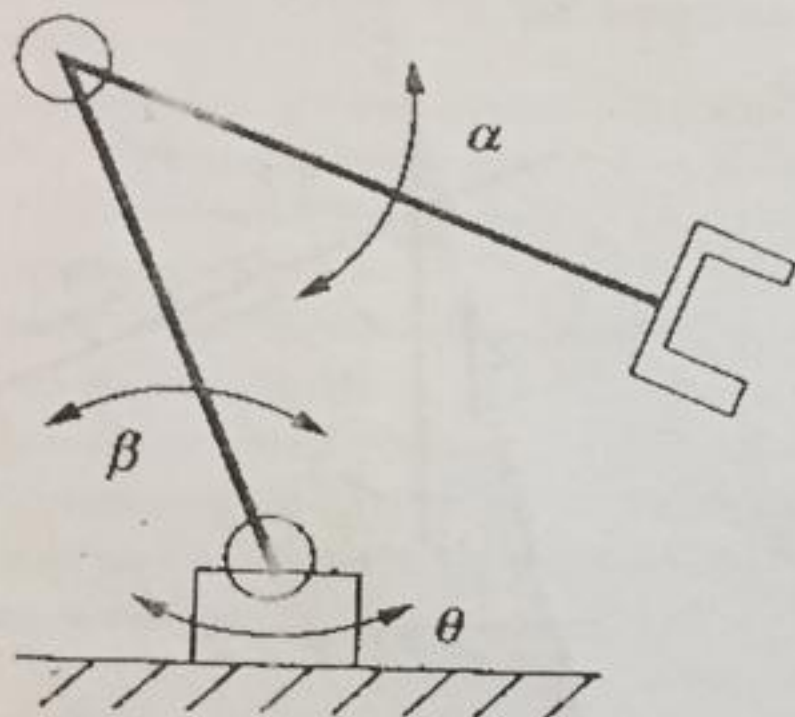
6. Jointed arm configuration:

Fig: Jointed arm configuration

- Fig shows Jointed arm configuration
- It is similar to the configuration of a human arm.
- It consists of several straight members connected by joints which are analogous to the human shoulder, elbow, and wrist and hence the name jointed arm configuration
- The vertical column is mounted on a base and that swivels about the base
- Wrist is attached to end of the arm and end effector is connected to the wrist
- Motions of this kind of robot traces Quasi-spherical work space (Quasi-spherical work space means semi spherical work shape)
- It is used for automobile spot welding or painting.

ADVANTAGES AND DISADVANTAGES OF ROBOTS

<u>ADVANTAGES OF ROBOTS</u>	<u>DISADVANTAGES OF ROBOTS</u>
- Performs multiple tasks - Performs repetitive tasks - Suitable for hazardous or uncomfortable working condition - Can operate for longer periods - Robots can perform the operations with a very high accuracy - Increases the quality of products	- Initial investment cost is high - Requires skilled programmer - Requires proper maintenance - Leads to unemployment problem - Lack of decision making power by robot

APPLICATIONS OF INDUSTRIAL ROBOTS

<u>GENERAL APPLICATIONS OF ROBOTS</u>	<u>PARTICULAR AREAS OF APPLICATIONS OF ROBOTS</u>
- Dangerous work environment for humans - Difficult handling task for humans - Repetitive Tasks - Boring and unpleasant tasks - Multi-shift Operation - Operating for long hours without rest - Dangerous work environment for humans	- Welding and spray painting - Drilling, Grinding, Polishing - Laser cutting - Water jet cutting - Loading and un-loading applications - Simple pick and place operations - Assembly task

»»»»END OF MODULE 3««««